Mastering Amazon Sales Forecasting with Machine Learning

This presentation outlines our final project: building and testing machine learning models using a cleaned Amazon Sales dataset. We'll explore the process from data preparation to advanced ensemble techniques, mirroring real-world data science applications.

From Raw Data to Refined Insights

In our mid-term project, we transformed raw data into a clean, usable format. This involved crucial steps that are foundational to any successful machine learning initiative.

Handling Missing Values

Strategically addressing gaps in the dataset to maintain data integrity.

Feature Engineering

Creating new, impactful features from existing data to enhance model performance.

Encoding Categorical Variables

Converting non-numeric data into a format suitable for machine learning algorithms.

Column Optimization

Removing irrelevant or redundant columns to streamline the dataset and reduce noise.

Real-World Impact: ML in E-commerce

Machine learning models are indispensable for e-commerce giants, driving strategic decisions that enhance efficiency and profitability. Our project replicates these critical business applications.

Predict Demand

Optimize inventory to avoid stockouts or overstocking.

Estimate Sales

Aid budgeting and financial planning with accurate forecasts.

Set Better Prices

Suggest optimal pricing strategies and discount timings.

Understand Customers

Identify best-selling items and anticipate buying trends.

Improve Recommendations

Personalize shopping experiences for higher conversion.

Optimize Marketing

Target promotions effectively to boost product visibility.

John's Contribution: Laying the Foundation

My focus was on establishing a robust modeling workflow and a clear performance baseline. I created the essential 70/15/15 train, validation, and test splits, which served as the data framework for the entire team.

I then trained a Linear Regression baseline model and meticulously calculated MAE, MSE, and R^2 for both validation and test sets. This provided a crucial first point of comparison for all subsequent, more complex models. This step was invaluable for understanding how preprocessing, feature setup, and data splitting integrate seamlessly.

Running the baseline deepened my comprehension of regression metrics and reinforced the importance of baselines before delving into advanced algorithms.

Saima's Contribution: Tree-Based Models & Visualization

My role involved developing Decision Tree and Random Forest regression models to forecast Amazon product purchases. The Decision Tree creates a detailed model by splitting data based on various features, while the Random Forest combines multiple trees to stabilize and enhance prediction accuracy.

I generated key visualizations: feature importance plots, actual vs. predicted scatter plots, and residual plots. Feature importance plots, especially from Random Forest, highlight which characteristics like ratings or discounts drive sales. Actual vs. predicted plots show prediction accuracy, with tighter clustering indicating better performance.

Residual plots reveal systematic errors, and Random Forest typically shows more random scattering around zero, indicating fewer consistent mistakes. These visualizations are vital for real-world e-commerce decisions, guiding product improvements and marketing strategies.

Misty's Contribution: Advanced Models & Hyperparameter Tuning

My task was to implement Gradient Boosting and K-Nearest Neighbors (KNN) regression, focusing on hyperparameter optimization to maximize their performance. Using GridSearchCV, I systematically searched for the best hyperparameters within predefined grids.

For Gradient Boosting, I fine-tuned parameters like the number of estimators, learning rate, and tree depth. For KNN, I optimized the number of neighbors and weighting scheme, all evaluated using cross-validation and negative mean squared error as the scoring metric. This rigorous tuning process allowed me to select optimal settings, fit the models to the training data, and generate predictions on the validation set.

Finally, I calculated MAE, MSE, and R^2 for both models, providing critical insights into their accuracy and goodness of fit, complementing the evaluations of Decision Tree and Random Forest.

Joseph's Contributions: Ensembles Methods

I implemented three ensemble methods: an average ensemble, a voting ensemble, and a Bayesian Ridge model. The average ensemble combined predictions from Random Forest, Gradient Boosting, and KNN. The voting ensemble used Linear Regression, Random Forest, and Gradient Boosting. The Bayesian Ridge model provided a probabilistic approach, balancing fit and uncertainty. The voting ensemble achieved the best validation R^2, while the Bayesian Ridge surprisingly showed the highest R^2 on the test set, demonstrating its strong generalization capabilities. This highlighted that different ensemble strategies excel under varying conditions.

Willie: Comprehensive Evaluation

My role was to meticulously evaluate and compare the performance of all five models—Linear Regression, Decision Tree, Random Forest, Gradient Boosting, and KNN (where applicable). I focused on Mean Absolute Error (MAE), Mean Squared Error (MSE), and R^2 Score, using a helper function to ensure consistent metric calculation across all predictions.

This systematic evaluation allowed us to identify the top-performing models based on their MAE, MSE, and R^2 scores. I also compiled all team contributions into a unified document, authored the "Best Performance" report, and the comprehensive "Full Approach" document, culminating our project's findings and methodology.